

Mathematical Tumor Model with Immune Resistance and Drug Therapy

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Introduction

- The goal of chemotherapy is to destroy the tumor cells, while maintaining adequate amounts of healthy tissue.
- If the therapy pushes the system into a limit cycle in which the size of the tumor is small for a long period of time (for example as long as the life of the patient), this could also be considered “a cure”.

Introduction

- The general goal is to keep the patient healthy while killing the tumor.
- We chose to minimize the tumor population, while constraining the normal cells to stay above some minimum level.

The Model: Overview

- $I(t)$: The number of immune cells at time t
- $T(t)$: The number of tumor cells at time t
- $N(t)$: the number of normal/host cells at time t

Initial Values

- $I(0) = \frac{s}{d_1}$
- $T(0) = 10^{-5}$
- $N(0) = 1$

The Model: Parameters

- The source of the immune cells is considered to be outside of the system, thus influx rate s
- In the absence of any tumor, the cells will die off at a per capita rate d_1
- ρ and α constant positives
- r_i and b_i growth rates and carrying capacities: $i = 1$ tumor cells, $i = 2$ normal cells

The Model: Equations

- Without using drugs: we would like the tumor-free equilibrium to be stable so that the possibility exists of moving the state of the system toward the tumor-free point.
- If the tumor free equilibrium is unstable, then according to this model, no amount of chemotherapy will be able to completely eradicate the tumor.

The Model: Equations

- Without using drug:

$$\dot{I} = s + \frac{\rho IT}{\alpha + T} - c_1 IT - d_1 I$$

$$\dot{T} = r_1 T(1 - b_1 T) - c_2 IT - c_3 TN$$

$$\dot{N} = r_2 N(1 - b_2 N) - c_4 TN$$

The Model: Equations

- With eigenvalues:

$$\lambda_1 = -d_1 < 0$$

$$\lambda_2 = r_1 - \frac{c_2 S}{d_1} - c_3$$

$$\lambda_3 = -r_2 < 0$$

- Tumor-free equilibrium is stable as long as $\lambda_2 < 0$ or:

$$r_1 < \frac{c_2 S}{d_1} + c_3$$

The Model: Equations

- Using drugs: We denote by $u(t)$ the amount of drug at the tumor site at time t . We assume that the drug kills all types of cells, but that the kill rate differs for each type of cell exponentially:

$$F(u) = a(1 - e^{-ku})$$

- Where $F(u)$ is the fraction cell kill for a given amount of drug u at the tumor site.

The Model: Equations

- Using drug:

$$\dot{I} = s + \frac{\rho IT}{\alpha + T} - c_1 IT - d_1 I - a_1(1 - e^{-u})I$$

$$\dot{T} = r_1 T(1 - b_1 T) - c_2 IT - c_3 TN - a_2(1 - e^{-u})T$$

$$\dot{N} = r_2 N(1 - b_2 N) - c_4 TN - a_3(1 - e^{-u})N$$

$$\dot{u} = v(t) - d_2 u$$

The Model: Equations

- Objective Function (to be minimized): $J(t_f) = T(t_f)$
- Constraint: $N(t) \geq 0.75, 0 \leq t \leq t_f$

Dynamic Programming

- States should be discrete, using:

$$x(k + 1) = x(k) + T_s f(k, x(k), u(k))$$

- Objective function should be discrete too, using:

$$J = h(x(N)) + \Delta t \sum_{k=0}^{N-1} g(x(k), u(k))$$

Dynamic Programming

- Discrete states:

$$I(k + 1) = I(k) + s + \frac{\rho I(k)T(k)}{\alpha + T(k)} - c_1 I(k)T(k) - d_1 I(k) - a_1(1 - e^{-u(k)})I(k)$$

$$T(k + 1) = T(k) + r_1 T(k)(1 - b_1 T(k)) - c_2 I(k)T(k) - c_3 T(k)N(k) - a_2(1 - e^{-u(k)})T(k)$$

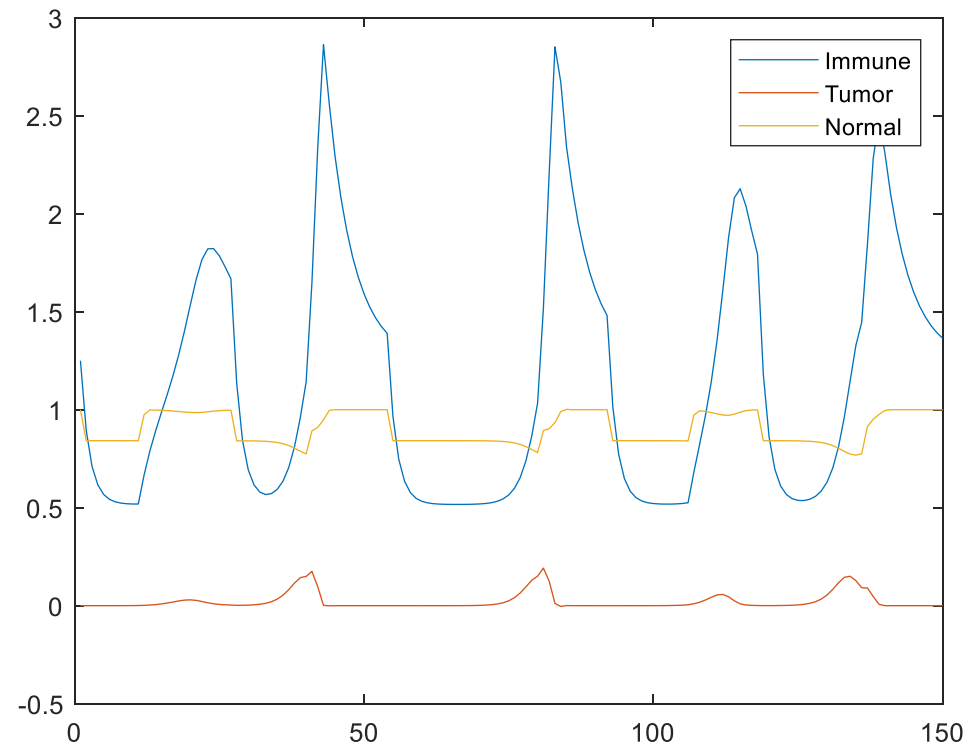
$$N(k + 1) = N(k) + r_2 N(k)(1 - b_2 N(k)) - c_4 T(k)N(k) - a_3(1 - e^{-u(k)})N(k)$$

$$u(k + 1) = v(k) + (1 - d_2)u(k)$$

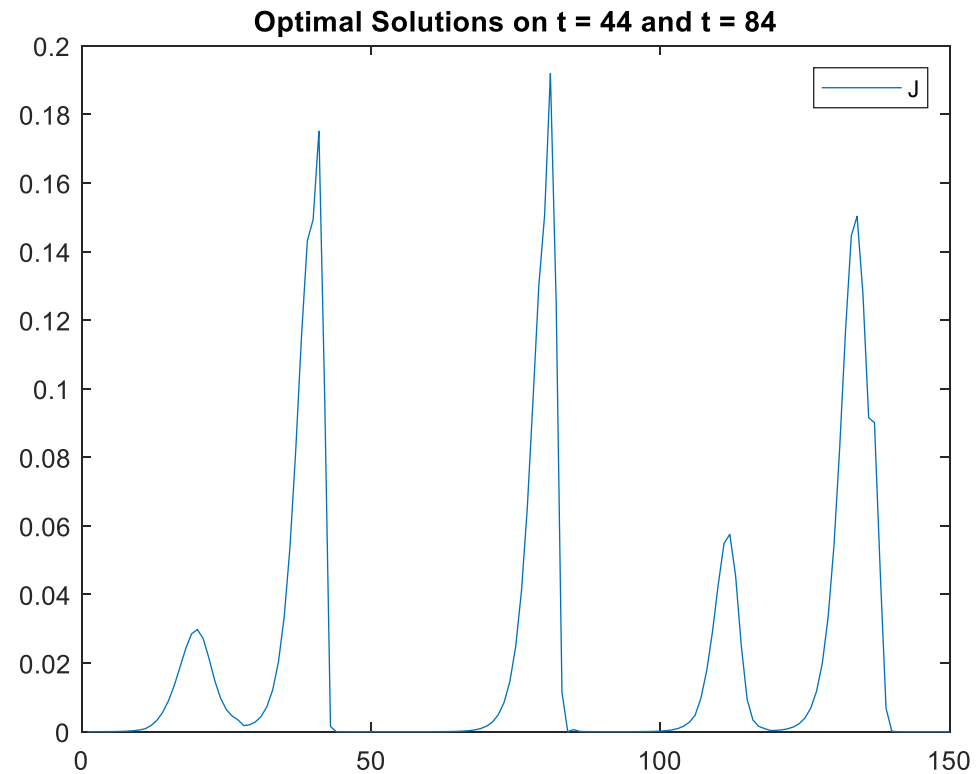
Dynamic Programming

- Continuous objective function: $J(t) = T(t)$
- Discrete objective function: $J(k) = T(k)$

Dynamic Programming



Dynamic Programming



Calculus of Variations

- Euler's first equation:

$$g_x - \frac{d}{dt} g_{\dot{x}} = 0$$

- Since our transfer function was $J(t_f) = T(t_f)$, we chose g to be $\dot{T}$, then:

$$J = \int_{t=0}^{t=t_f} \dot{T} dt$$

Calculus of Variations

- We know t is always $0 \leq t \leq t_f$
- $x(0) = T(0) = 10^{-5}$
- t_f and $T(t_f)$ free, so we got two new constraints besides the Euler's first equation:

$$g(x^*(t_f), \dot{x}^*(t_f), t_f) = 0$$

$$g_{\dot{x}}(x^*(t_f), \dot{x}^*(t_f), t_f) = 0$$

Calculus of Variations

- We determine g_x :

$$g_x = \begin{bmatrix} g_I \\ g_T \\ g_N \end{bmatrix} = \begin{bmatrix} \frac{\partial \dot{T}}{\partial I} \\ \frac{\partial \dot{T}}{\partial T} \\ \frac{\partial \dot{T}}{\partial N} \end{bmatrix} = \begin{bmatrix} -c_2 T \\ r_1(1 - b_1 T) - r_1 b_1 T - c_2 I - c_3 N - a_2(1 - e^{-u}) \\ -c_3 T \end{bmatrix}$$

Calculus of Variations

- By solving $g_x = 0$ we conclude:

$$T^* = 0$$

$$r_1 - c_2 I - c_3 N - a_2(1 - e^{-u}) = 0$$

Calculus of Variations

- Determining $g_{\dot{x}}$:

$$g_{\dot{x}} = \begin{bmatrix} g_{\dot{i}} \\ g_{\dot{T}} \\ g_{\dot{N}} \end{bmatrix} = \begin{bmatrix} \frac{\partial \dot{T}}{\partial \dot{i}} \\ \frac{\partial \dot{T}}{\partial \dot{T}} \\ \frac{\partial \dot{T}}{\partial \dot{N}} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Calculus of Variations

- Since $g_{\dot{x}}$ could never be equal to zero, we could not use the calculus of variations to solve this optimization problem.

Hamiltonian Method

- The differential equation is as follows: $h(I, T, N, u, v) = (\dot{I}, \dot{T}, \dot{N}, \dot{u})$
- For simplicity, we use $(I, T, N, u) = (x_1, x_2, x_3, x_4)$
- Objective function is redefined as: $J(x, v) = T(t_f) = x_2(t_f)$
- Inequality constraint: $k(x, t, v) = x_3(t) - 0.75 \geq 0, 0 \leq t \leq t_f$

Hamiltonian Method

- Hamiltonian for this optimization problem:

$$H = p_1(x_1) + p_2(x_2) + p_3(x_3) + p_4(x_4) + \eta k$$

- We could define $p_i(t)$ as it follows:

$$\dot{p}_i = -\frac{\partial H}{\partial x_i}$$

Hamiltonian Method

$$\dot{p}_1 = -p_1 \left(\frac{\rho x_2}{\alpha + x_2} - c_1 x_2 - d_1 - a_1(1 - e^{-x_4}) \right) + p_2 c_2 x_2$$

$$\dot{p}_2 = -p_1 \left(\frac{\rho \alpha x_1}{(\alpha + x_2)^2} - c_1 x_1 \right) - p_2 (r_1 - 2r_1 b_1 x_2 - c_2 x_1 - c_3 x_3 - a_2(1 - e^{-x_4})) + p_3 c_4 x_3$$

$$\dot{p}_3 = p_2 c_3 x_3 - p_3 (r_2 - 2r_2 x_3^2 - c_4 x_2 - a_3(1 - e^{-x_4})) - \eta(t)$$

$$\dot{p}_4 = -e^{-x_4} (a_1 p_1 x_1 + a_2 p_2 x_2 + a_3 p_3 x_3)$$

Hamiltonian Method

- Combining $\eta(t) = 0, \eta(t)k(t) = 0$, and the inequality constraint:

$$\eta(t) = \begin{cases} 1 & \text{for } x_3(t) \leq 0.75 \\ 0 & \text{otherwise} \end{cases}$$

- $p_1(t_f) = 0, p_2(t_f) = 1, p_3(t_f) = 0, p_4(t_f) = 0$

Hamiltonian Method

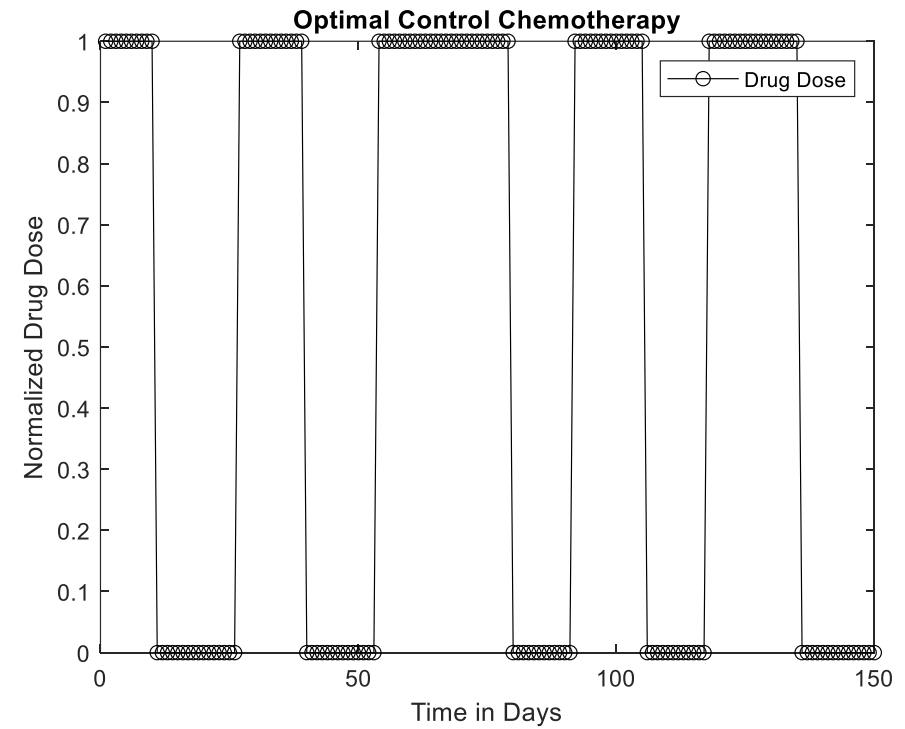
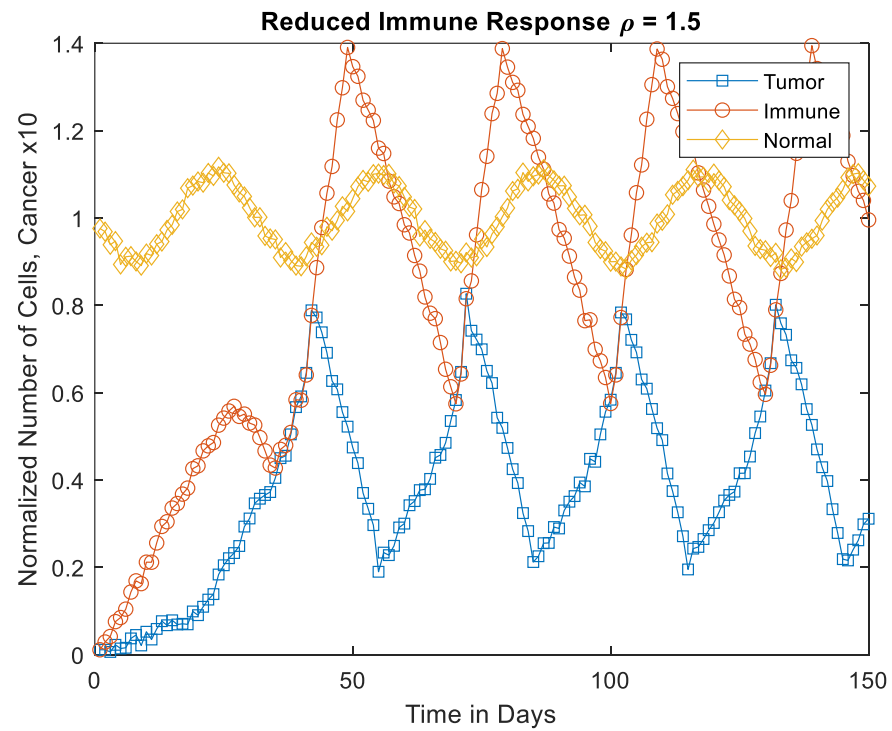
- Control equation: $\frac{\partial H}{\partial v} = p_4$
- v is defined:

$$v = \begin{cases} 0 & p_4 > 0 \\ v_{max} & p_4 < 0 \\ singular & p_4 = 0 \end{cases}$$

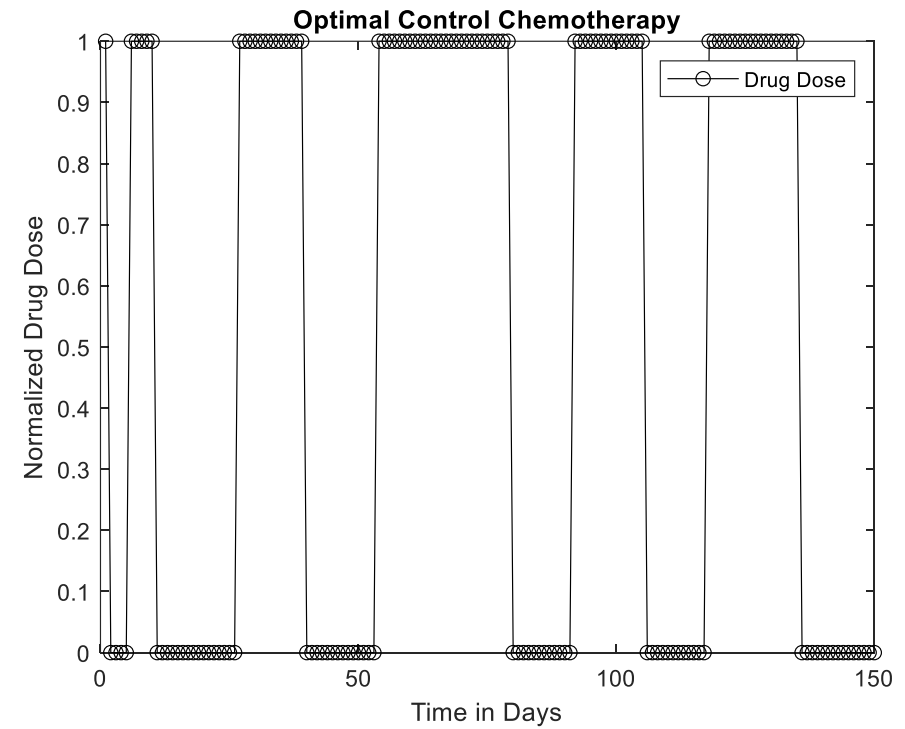
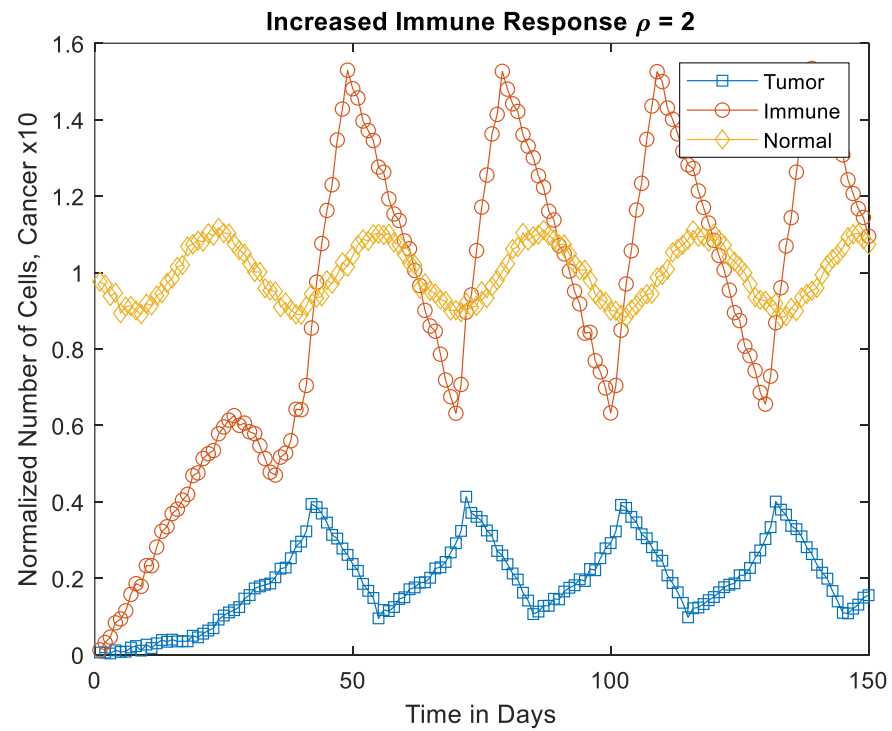
Used Parameters

- Fraction Cell Kill: $0 \leq a_i \leq 0.5$, and $a_3 \leq a_1 \leq a_2$
- Carrying Capacities: $b_1^{-1} \leq b_2^{-1} = 1$
- Competition Terms: $c_1, c_2, c_3, c_4 > 0$, and c_2 bigger than the rest
- Death Rates: $d_1 = 0.2$ death rate of immune cells, $d_2 = 1$ death rate of drug
- Per Unit Growth Rates: $r_2 = 1$, and $r_1 > r_2$
- Immune Source Rate: $0 \leq s \leq 0.5$
- Immune Threshold Rate: $\alpha = 0.3$
- Immune Response Rate: ρ at least 1, and could be (0,2.5)

Simulation Results



Simulation Results



Reference

- 1. A Mathematical Tumor Model with Immune Resistance and Drug Therapy: an Optimal Control Approach**, L.G. DePillis, and A. Radunskaya, *Journal of Theoretical Medicine*, Vol. 3, pp. 79-100, April 18, 2000.

Thank You

